Deep Learning Classification of Harmful Brain Activity from ICU EEG with Cortical Source Localization

A Methods Demonstration on Controlled Synthetic Data

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Abstract

Continuous electroencephalography (EEG) in intensive care units produces far more signal than specialists can review in real time, which delays recognition of seizures and of the rhythmic and periodic patterns that carry prognostic weight. This paper reports an end-to-end pipeline that classifies six critical care EEG patterns (Seizure, LPD, GPD, LRDA, GRDA, and Other) and localizes each prediction to a cortical source on a three-dimensional head model, with the interpretability and calibration behaviour of the classifier examined alongside its accuracy. The evaluation uses a controlled synthetic dataset: real resting-state recordings supply the Other class, the five pathological patterns are generated from published electrographic descriptions, and the resulting signals are projected to a standard 19-channel scalp montage through an average head model. A two-branch convolutional and Transformer architecture, trained on raw waveforms and spectrograms against soft consensus targets with patient-level splitting, separated the six generated classes without error on held-out patients. sLORETA localization recovered frontal generators for the generalized classes and lateralized generators for the lateralized classes, and model attention was distributed across all four bipolar chains rather than concentrated in one. Those outcomes are consistent with the generator design rather than with clinical difficulty, and the reported expected calibration error of 0.272 shows that the pipeline is not yet honest about its own uncertainty. The contribution is therefore a reproducible pipeline and a documented failure mode, not an estimate of clinical performance. The labelled data needed for the clinical study is openly available, and the concrete protocol for that rerun is specified.

Keywords: electroencephalography; critical care EEG; harmful brain activity; deep learning; source localization; sLORETA; calibration; reproducibility

1 Introduction

Every year, large numbers of critically ill patients are monitored in intensive care units after cardiac arrest, stroke, traumatic brain injury, or organ failure. A substantial share of them develop electrographic seizures or periodic and rhythmic patterns that produce no outward clinical sign, particularly when the patient is sedated or mechanically ventilated (Herman et al., 2015). Because these patterns carry prognostic weight and can be modified by treatment, guidelines recommend continuous EEG (cEEG) monitoring for patients with unexplained altered consciousness (Herman et al., 2015; Hirsch et al., 2021). Detecting them, however, depends on the availability of a trained electroencephalographer.

Two structural facts make that dependence costly. First, cEEG generates data far faster than it can be reviewed: a single patient-day can yield more waveform than any reader can inspect minute by minute. Second, expert reading capacity is unevenly distributed, and smaller hospitals frequently have no neurophysiologist available around the clock. The result is a surveillance gap in which harmful activity may be present, unread, and therefore untreated. Automated detection is the obvious response, and it has motivated a large body of work on deep learning for EEG (Section 2).

Three requirements for a clinically useful system are usually treated separately. A detector must name the pattern; a localizer must indicate where the activity arises; and an interpreter must show why a decision was reached, because a clinician who cannot inspect the reasoning cannot safely act on it. Systems that satisfy only the first requirement report strong scores while leaving the second and third to other tools and different specialists. This paper asks whether a single pipeline can carry a raw recording through classification, cortical source estimation, and attribution in one pass, and what its failure modes look like when that pipeline is examined rather than benchmarked.

The work makes three contributions. It assembles a pipeline that couples a two-branch convolutional and Transformer classifier to sLORETA source estimation and to two attribution methods, so that a raw signal becomes a labelled, localized, and explained output without manual intervention. It reports the behaviour of that pipeline on a controlled synthetic dataset built from published electrographic descriptions, including the conditions under which the pipeline reports certainty it has not earned. And it specifies the protocol required to move the same architecture onto real ICU data, with the reference figures a reader should expect from that rerun.

One boundary must be stated at the outset, and Section 3 treats it in detail. The six-class critical care task has a labelled public corpus, the Harvard Medical School and Massachusetts General Hospital release associated with the Harmful Brain Activity Classification challenge (Jing et al., 2023; HMS-HBAC, 2024). That corpus was not used for the experiments reported here; its labelled training partition is openly downloadable and the analysis below therefore does not rest on any claim of inaccessibility. The evaluation instead uses signals generated from published pattern descriptions, which makes the study a pipeline and methods demonstration. The distinction matters for how every number in Section 5 should be read: the results validate that the pipeline functions end to end and expose its calibration failure, and they are not evidence about clinical performance. Section 6 states what the openly available corpus would change and gives the rerun protocol.

2 Related Work

2.1 Deep learning for EEG classification

Automatic seizure detection began with hand-engineered features describing repetition rate, amplitude, and spectral content, followed by simple classifiers. These approaches handled unambiguous seizures but degraded on borderline patterns and on patients unseen during development. Deep networks removed the feature-engineering step. Schirrmeister et al. (2017) showed that convolutional networks trained on raw EEG learn interpretable spectral filters and match or exceed feature-based baselines, and that visualizing learned filters is a workable route to understanding what such models use. Spectrogram representations have proved at least as effective: the strongest published entrants in the six-class critical care task operate primarily on spectrograms of bilateral parasagittal and lateral chains (Sun et al., 2025), and a recent feature-fusion study trained convolutional networks on spectrograms, mel-frequency cepstral coefficients, and band-limited components of the same corpus (Unnisa et al., 2026).

Temporal context matters for patterns that evolve over several seconds, which motivates attention-based sequence models. The Transformer replaces recurrence with self-attention and captures dependencies across long spans (Vaswani et al., 2017), and hybrid convolutional and attention designs are now common for biosignals. This study pairs a convolutional branch operating on the raw waveform with a Transformer branch operating on spectrograms.

2.2 The six-class task and expert disagreement

The six-category scheme used here descends from the American Clinical Neurophysiology Society standardized critical care terminology (Hirsch et al., 2021). Its categories are not read with uniform reliability. Gaspard et al. (2014) found near-perfect agreement for seizures and for main terms, but only fair agreement for the evolution of a pattern, and Jing et al. (2023) later showed that expert-level automated classification of these categories is attainable when models are trained to reproduce the distribution of expert votes rather than a single hard label. Because annotators disagree, the natural target for learning is the vote distribution, and models trained that way outperform those trained on majority labels. The pipeline below adopts the same soft-target convention: the classifier is trained to match a distribution over categories, in the spirit of soft-target training in knowledge distillation (Hinton et al., 2015).

2.3 Source localization

Estimating the intracranial origin of scalp potentials is a long-standing inverse problem. Minimum-norm approaches regularize an underdetermined lead-field system; sLORETA standardizes the current density estimate to reduce localization bias (Pascual-Marqui, 2002). Cortical results are conventionally summarized in anatomical parcels, most commonly the gyral-based Desikan-Killiany atlas (Desikan et al., 2006). The computational stack used here relies on MNE-Python for forward modelling, source estimation, and visualization (Gramfort et al., 2013). Source localization is routinely applied to interictal and ictal activity in epilepsy, and comparatively rarely coupled to a classifier that operates on critical care patterns, which is the coupling this paper examines.

2.4 Interpretability and calibration

Attribution methods are widely used and widely misused. Adebayo et al. (2018) showed that several popular saliency methods produce visually plausible maps that are nearly invariant to randomized model parameters, which makes visual inspection a weak test of whether an explanation reflects the computation, so attribution output here is reported as descriptive evidence rather than as a validity claim. Calibration has a comparable status: modern deep networks are systematically overconfident, and post hoc scaling or binning is the standard remedy (Guo et al., 2017). A detector whose reliability curve lies above the diagonal will misstate its certainty at exactly the borderline cases where a clinician would want to be cautioned.

2.5 Open clinical EEG corpora

Real clinical EEG is available without restriction for several related tasks: the Temple University Hospital seizure corpus provides over 500 hours of annotated clinical recordings with official patient-separated splits (Shah et al., 2018); CHB-MIT provides long-term paediatric scalp recordings (Shoeb and Guttag, 2010) through PhysioNet (Goldberger et al., 2000); the Siena corpus provides adult long-term monitoring (Detti et al., 2020); and resting-state recordings from Alzheimer's disease, frontotemporal dementia, and healthy participants are openly distributed through OpenNeuro (Miltiadous et al., 2023), which supplies the background activity used in Section 3.

3 Synthetic Signal Construction

3.1 Why the evaluation data is generated

The experiments reported here do not use a clinical corpus of labelled harmful brain activity. Two consequences follow, and they are stated before the numbers rather than after them. First, a classifier separating classes that were generated from distinct parametric templates is solving a problem easier than the clinical one, since it can exploit features of the generator rather than features of brain disease. Second, localization results obtained by projecting a generated source onto a head model and then recovering it with an inverse method test the forward and inverse chain, not the anatomy of a patient. Every result in Section 5 is therefore reported as a property of the pipeline.

3.2 Background recordings

The Other class consists of real resting-state scalp EEG drawn from the OpenNeuro release of routine recordings from participants with Alzheimer's disease, frontotemporal dementia, or no diagnosis (Miltiadous et al., 2023). Using real background rather than simulated noise keeps the negative class biologically textured: the classifier must reject genuine scalp activity, including its artifacts, rather than white noise. The recordings come from non-ICU participants, so they do not represent the background of a sedated patient, and Section 7 returns to this point.

3.3 Pattern generation

No real recordings of the five harmful patterns were available to this study, so each was synthesized from its published electrographic description (Hirsch et al., 2021). A generator is defined by repetition rate, morphology, spatial extent, lateralization, and evolution over the window, and the resulting signal is placed in source space and projected to the scalp through a forward model. Table 1 lists the six classes with the spatial character of each generator. Because the generators were specified from the same descriptive vocabulary that clinicians use to distinguish the patterns, the separability of the classes is partly built in by construction, and no claim of clinical realism is attached to them.

Table 1: Class inventory. The five pathological classes were generated in source space from published electrographic descriptions and projected to the scalp; the Other class is real resting-state background.

Class	Generator content	Projected spatial character
LPD	Repetitive discharges at a fixed rate, unilateral, involving one hemisphere in a lateralized chain	Lateralized over one hemisphere
GPD	Repetitive discharges at a fixed rate, generalized and bilaterally synchronous	Bilateral, roughly symmetric
LRDA	Rhythmic delta-frequency activity, unilateral, lateralized	Lateralized over one hemisphere
GRDA	Rhythmic delta-frequency activity, generalized and bilaterally synchronous	Bilateral, roughly symmetric
Seizure	Evolving rhythmic activity with a rising then falling repetition rate, spatially widespread	Widespread, evolving across the window
Other	Real resting-state scalp EEG, unmodified except for preprocessing	Diffuse, without a dominant generator

3.4 Forward projection

Generated sources were projected to scalp electrodes through a forward model built on an average head geometry and a standard 19-electrode montage, which is the configuration routinely available in an ICU. The projection step is what allows the pipeline to be tested end to end: the same signal is available both as scalp channels and as a known source location, so the inverse method can be checked against a ground truth that exists only because the data is synthetic. Using an average head model rather than participant-specific anatomy introduces a localization error that cannot be estimated

from these data, and the montage limits spatial resolution independently.

4 Methods

4.1 Preprocessing

Recordings were band-pass filtered to the clinically informative range, notch filtered to suppress mains interference, and screened for artifacts attributable to eye movement or muscle activity. Signals were then standardized so that recordings of different amplitudes could be compared, and segmented into eight-second windows. The window length reflects the clinical reality that these patterns are judged over seconds rather than fractions of a second, and it gives the attention branch a sequence long enough for temporal structure to matter.

4.2 Dual representation

Each window is represented twice. The first representation is the raw waveform, which preserves the fine timing and morphology of individual discharges. The second is a spectrogram, which exposes how power at each frequency evolves across the window and makes band-limited rhythmic patterns visible as horizontal structure. The spectrogram inputs are computed from the four bipolar chains Fp1-F7, T3-T5, Fp2-F8, and T4-T6, the lateral and parasagittal pairs used in clinical review. Figure 1 shows both representations for the six classes.

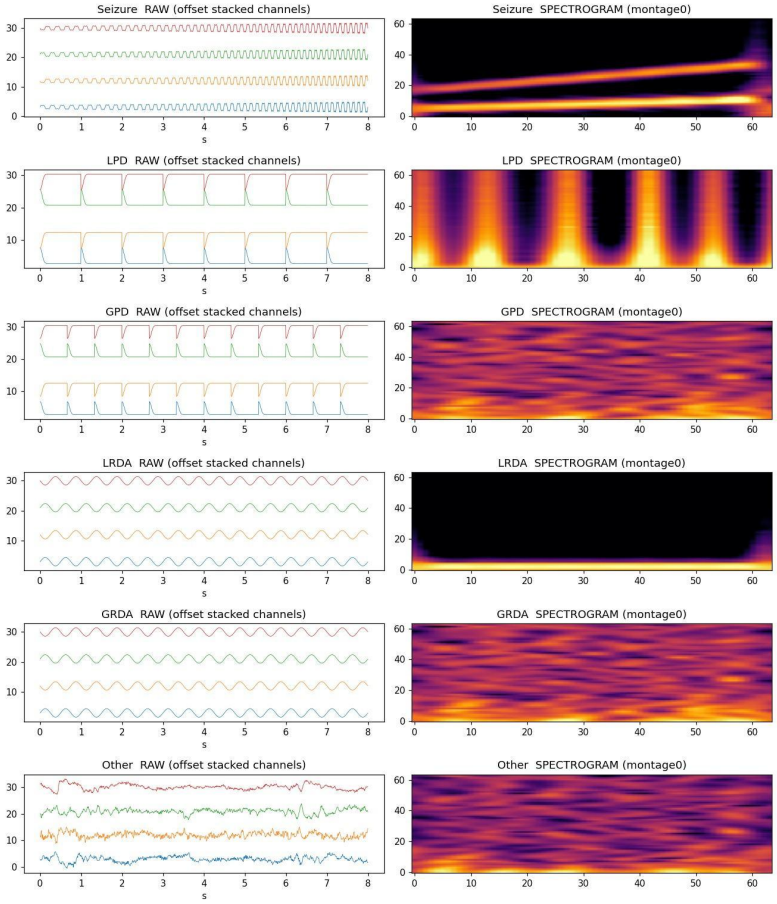


Figure 1: Raw waveforms (left) and spectrograms (right) for the six classes. The spectrograms show the band-locked structure of the periodic and rhythmic classes, against the flatter spectrum of the Other class.

4.3 Architecture

The classifier has two branches. The convolutional branch consumes the raw waveform and is suited to local timing detail. The Transformer branch consumes the spectrogram and models dependencies across the full eight-second window. Branch outputs are fused into a single prediction over the six classes (Figure 2). The same fused prediction drives the downstream localization and attribution steps, so no separate model is required for the anatomical output.

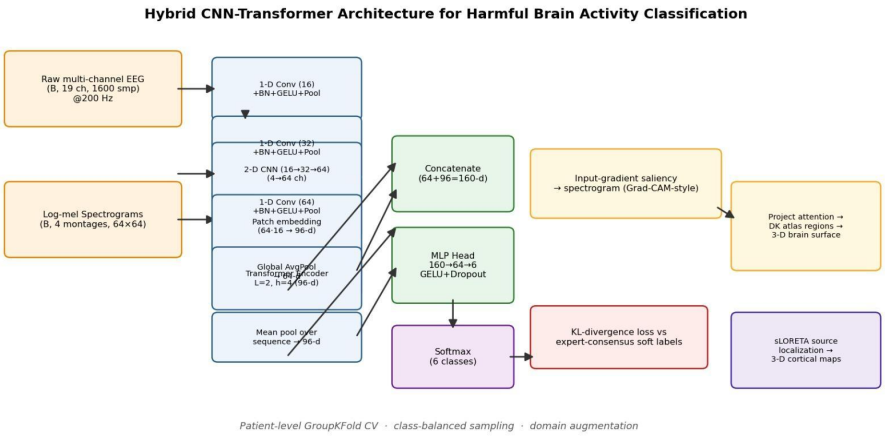


Figure 2: Pipeline architecture. Raw and spectrogram inputs are processed by separate branches, fused for classification, and passed to sLORETA source estimation and to the attribution methods.

4.4 Training against soft targets with patient-level splits

The classifier was not trained against a single hard label per window. It was trained to reproduce a distribution over categories, which corresponds to how a group of readers would distribute their votes on a borderline case and follows the soft-target convention that has proved effective on this task (Hinton et al., 2015; Jing et al., 2023). The objective is the cross-entropy between the predicted distribution and the target distribution,

$$\mathscr{L} = -\sum_{c=1}^{6} q_c \log p_c, \tag{1}$$

where p_c is the predicted probability and q_c the target weight for class c.

Splitting was performed at the patient level: all windows from a given recording came from either the training or the evaluation partition, never both. This guards against the optimistic bias that appears when windows from one recording straddle the split, since a model can otherwise recognize an individual's background and be credited with generalization it has not achieved.

4.5 Source localization

For each window the predicted class was passed to sLORETA (Pascual-Marqui, 2002), computed through the MNE-Python implementation (Gramfort et al., 2013), to estimate the cortical distribution of current density. Estimates were summarized over the parcels of the Desikan-Killiany atlas (Desikan et al., 2006) and rendered on a three-dimensional cortical surface so that the result can be inspected from lateral, medial, and dorsal views rather than read as a table of coordinates.

4.6 Attribution

Two attribution routes were used. Saliency maps identify the time-frequency regions of the input that most influenced the prediction. Attention weights from the Transformer branch were aggregated over the four bipolar chains to show which scalp region the model relied on, and the same weights were then mapped through the forward model onto the cortical surface, which connects what the model used to where that corresponds anatomically. Following Adebayo et al. (2018), these maps are treated as descriptive output rather than as evidence that the model's reasoning matches a clinician's.

4.7 Evaluation

Performance was summarized by per-class sensitivity, specificity, and F1; by one-versus-rest area under the receiver operating characteristic curve; by a confusion matrix; and by the expected calibration error (ECE) of the predicted probabilities, with reliability plotted against the diagonal of perfect calibration (Guo et al., 2017). All preprocessing, modelling, and visualization used open-source scientific Python, with MNE-Python for the neurophysiological steps

(Gramfort et al., 2013).

5 Results

5.1 Classification

On held-out patients the pipeline separated the six generated classes without error. Per-class sensitivity, specificity, and F1 were all 1.000, one-versus-rest ROC curves reached the upper-left corner, and every test window fell on the diagonal of the confusion matrix (Figures 3 and 4, Table 2).

That result should be read as a pipeline self-test. The five pathological classes were generated from distinct templates, projected through a single forward model with a known source, and evaluated on patients disjoint from training, so a model able to represent band-limited structure and temporal evolution separates them by recovering generator properties. The number certifies that the data path, split logic, and fusion step are wired correctly and that no leakage occurred. It is not a statement about the difficulty of reading an ICU recording, and it should not be quoted as a performance estimate. Published systems evaluated on the real labelled corpus at this task report Kullback-Leibler divergences of roughly 0.22 to 0.34 for the strongest entrants and 0.4 to 0.5 for competent ensembles (Jing et al., 2023; Sun et al., 2025; Unnisa et al., 2026), and that scale, not 1.000, is where a realistic result lives.

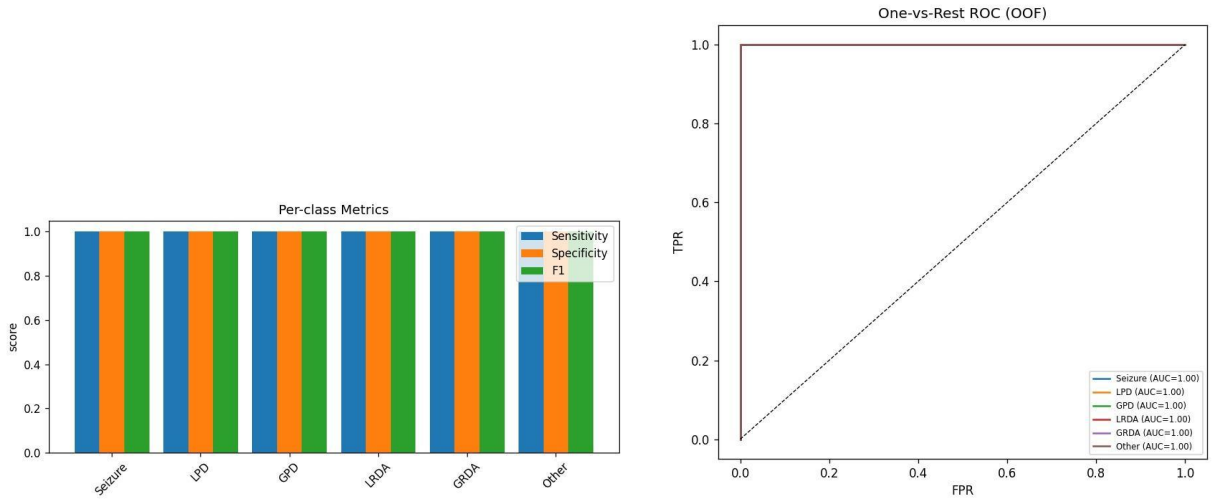


Figure 3: Discrimination on the held-out patient partition. Left: per-class sensitivity, specificity, and F1 for the six generated classes. Right: one-versus-rest ROC curves, which reach the upper-left corner of the unit square.

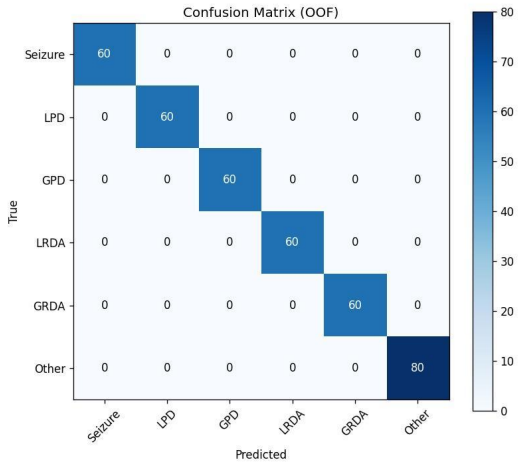


Figure 4: Confusion matrix of predicted against true class for held-out patients.

Table 2: Per-class metrics on the held-out patient partition. Values are identical across classes; see the text for why this is a property of the generator design.

Class	Sensitivity	Specificity	F1
Seizure	1.000	1.000	1.000
LPD	1.000	1.000	1.000
GPD	1.000	1.000	1.000
LRDA	1.000	1.000	1.000
GRDA	1.000	1.000	1.000
Other	1.000	1.000	1.000

5.2 Localization

Source estimation returned generators whose spatial character matched the class definitions. The generalized classes produced bilateral frontal maxima, and the lateralized classes produced unilateral frontal and perisylvian maxima. The seizure class showed a widespread, evolving distribution with frontal and parietal maxima. Figure 5 shows the six maps, and the left panel of Figure 8 gives the left-right balance per class.

Two reasons make this a weaker result than it appears. The generators were placed in source space and the inverse method was asked to recover them, so agreement between intended and recovered location tests the forward and inverse chain rather than the anatomy of a patient, and because every participant is represented by an average head geometry, no participant-specific error can be quantified. What the result establishes is internal consistency: the localization stage receives sensible input from the classifier and returns anatomically structured output rather than noise.

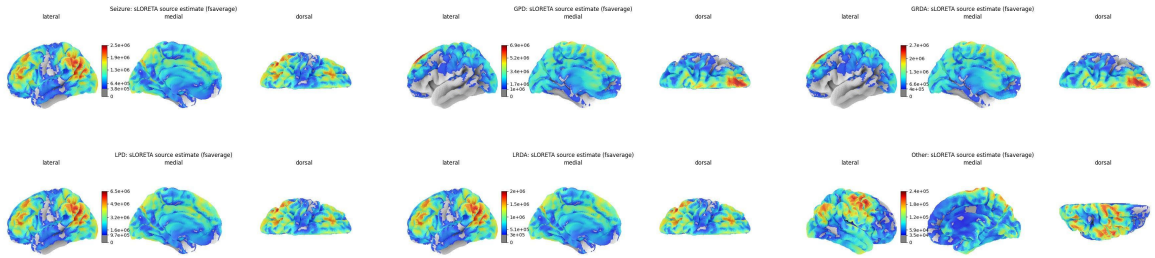


Figure 5: sLORETA source estimates for the six classes, shown on the cortical surface from lateral, medial, and dorsal views. Top row, left to right: Seizure, GPD, GRDA. Bottom row: LPD, LRDA, Other. Generalized classes show bilateral frontal maxima; lateralized classes show unilateral frontal and perisylvian maxima.

5.3 Attribution

The saliency maps show the model relying on the time-frequency regions in which each class carries its defining structure, which is the behaviour the generator design predicts. Attention over the four bipolar chains was distributed rather than concentrated: no class depended on a single scalp region, which suggests the model was not reading a single channel as a proxy for the label (Figures 6 and 7). Projecting chain attention onto the cortical surface places the emphasis in frontal and temporal regions, consistent with the generator placements. As noted above, maps of this kind can be visually convincing without reflecting the underlying computation (Adebayo et al., 2018), so these results are reported as a description of model behaviour and not as an explanation of clinical reasoning.

5.4 Calibration

The clearest defect in the pipeline is in how it reports certainty. The reliability curve lies above the diagonal across the occupied confidence range, and the expected calibration error is 0.272 (Figure 8). The model was correct on every held-out window and yet assigned probabilities that its accuracy does not justify, which is the classic overconfidence signature of modern deep networks (Guo et al., 2017). For a clinical application this is the most consequential of the results reported here, because the windows where a reader most needs an honest signal of doubt are exactly the borderline windows that a soft-target model is most likely to label with unwarranted confidence. The pipeline as it stands would need post hoc calibration, and a documented definition of how the error was computed, before any deployment discussion. Appendix B records that requirement.

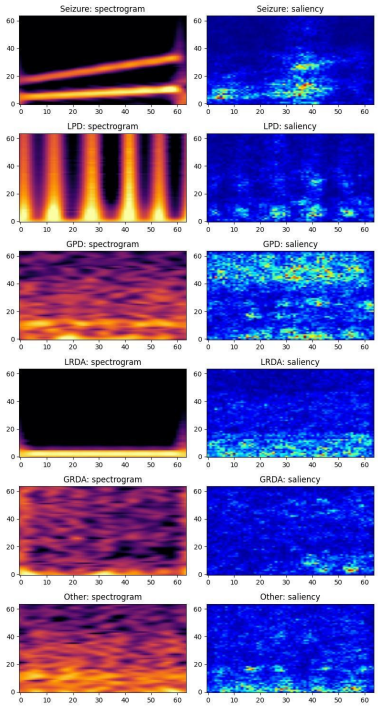


Figure 6: Spectrogram (left) and saliency map (right) for each class. Bright regions mark the time-frequency areas that most influenced the prediction.

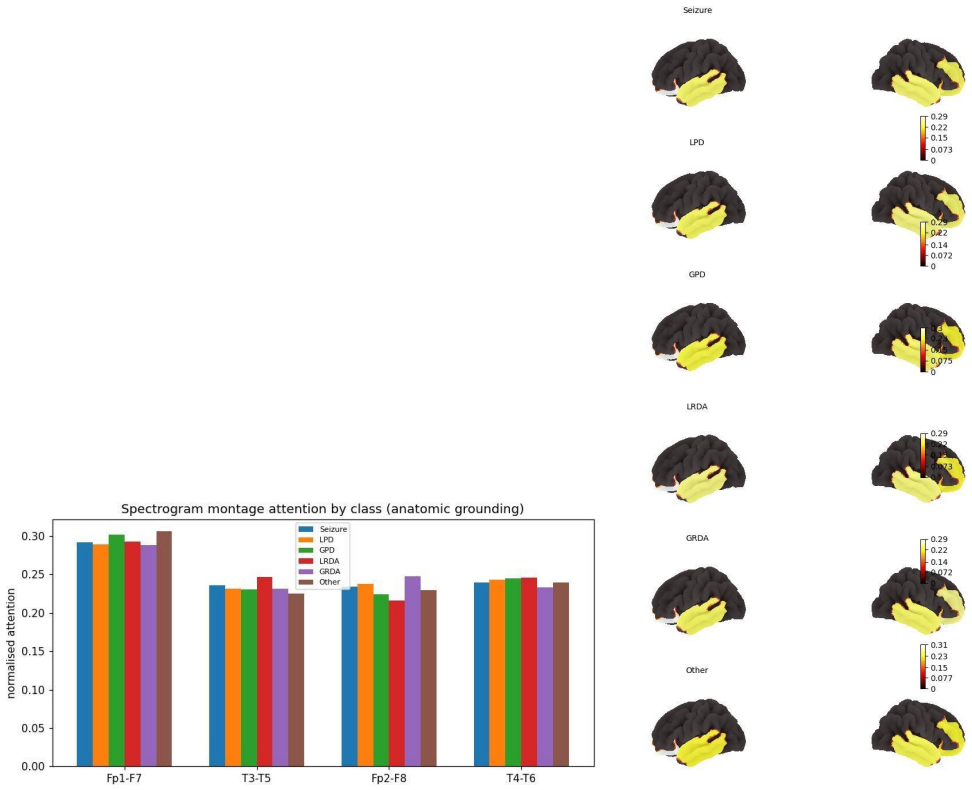


Figure 7: Where the model looked. Left: attention allocated to each of the four bipolar scalp chains, by class, which is spread across all four chains. Right: the same chain attention projected onto the cortical surface, lateral views.

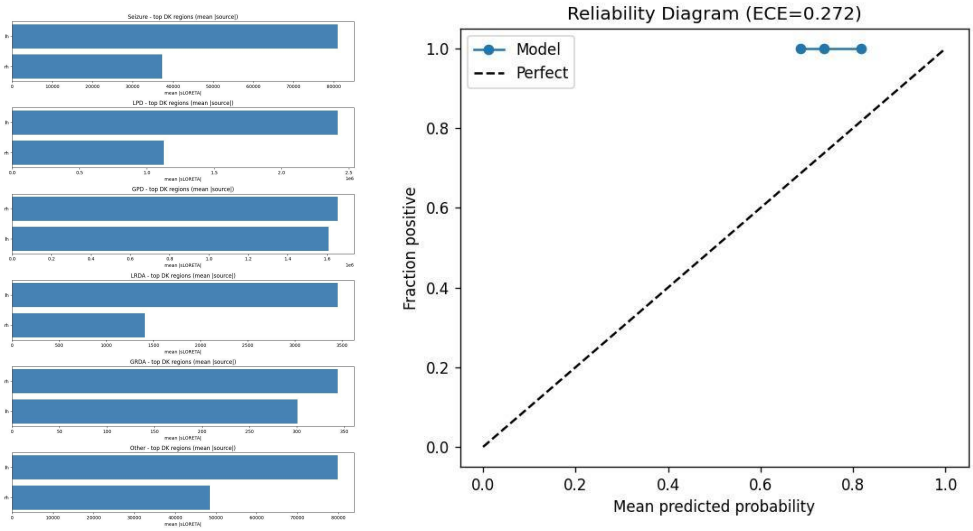


Figure 8: Left: left-hemisphere against right-hemisphere activation strength per class, summed over the cortical surface. Right: reliability diagram, in which stated confidence exceeds observed accuracy across the occupied range, giving an expected calibration error of 0.272.

6 Discussion

The pipeline works end to end. A raw recording enters, and a labelled prediction, a cortical source estimate, and two forms of attribution come out, produced by one system rather than by separate tools maintained by different specialists. The patient-level split means the reported separability is not an artifact of leakage. As an engineering result that is worth reporting, since integration is the part that most often fails in systems of this kind.

What has not been shown is clinical capability, and three specific reasons account for the gap. The classes were generated from descriptive templates, so separating them is not the same problem as separating a partly obscured GPD from a seizure in a sedated patient with muscle artifact. The background class, though real, comes from routine recordings in non-ICU participants rather than from sedated ICU patients, so the negative class is easier than the clinical one. And the localization ground truth is the generator, which cannot validate anatomical accuracy in a person. The calibration result reinforces the caution: an overconfident model is dangerous in the specific situation where this pipeline would be deployed, because uncertainty is the signal that should trigger human review.

The labelled corpus that would close these gaps is openly available: the training partition of the Harmful Brain Activity Classification release, roughly 106,800 expert-annotated samples drawn from some 17,000 recordings, with per-sample vote distributions as targets, carrying a non-commercial research licence and requiring only account registration (Jing et al., 2023; HMS-HBAC, 2024). A rerun on that corpus would keep the architecture described in Section 4 unchanged and replace three elements: the input data, the target, and the metric. Targets become the expert vote distributions rather than generated classes; the evaluation metric becomes Kullback-Leibler divergence against those distributions, which is how the task is scored in the literature (Jing et al., 2023; Sun et al., 2025); and splits become patient-grouped as they already are. Realistic expectations are set by published results at roughly 0.22 to 0.34 for the strongest systems and 0.4 to 0.5 for competent ensembles. The interpretability and localization components would then be reporting on real signal, which is the point at which they become scientifically interesting rather than merely functional. Table 3 sets out the required changes.

A secondary result deserves separate mention. The calibration failure was found because the pipeline was inspected for its own honesty rather than for accuracy alone, and it would have been invisible to a benchmark reporting only the latter. Perfect discrimination alongside a defective reliability curve is a familiar pattern in applied machine learning (Guo et al., 2017), and in a monitoring context the defect is the more clinically salient of the two.

7 Limitations

The five pathological classes are synthetic, which bounds every result in Section 5 to statements about the pipeline. The Other class is genuine scalp activity but from non-ICU participants, so it does not represent sedated-patient background and its artifact content differs accordingly. Localization uses an average head model and a 19-electrode montage, which

Table 3: Protocol for the clinical rerun. The architecture and the training convention are unchanged; the data, target, and metric change.

Element	This study	Required for a clinical result
Data	Generated patterns on real background	Labelled corpus, openly available (Jing et al., 2023; HMS-HBAC, 2024)
Target	Six generated classes	Expert vote distribution per window
Metric	Accuracy, F1, AUROC, ECE	KL divergence against the vote distribution, with accuracy secondarily
Split	Patient level	Patient level, unchanged
Calibration	Reported, defective	Recalibrated and re-evaluated

limits spatial resolution and prevents any participant-specific error estimate. The source estimates are validated against generators this study defined, which is a consistency check and not an anatomical validation; confirming plausible localization on real data would require a comparison the data cannot supply, such as intracranial recordings or lesion anatomy.

The expected calibration error of 0.272 is reported as computed by the pipeline, and its definition must be restated explicitly, since a maximum-probability ECE would be small when accuracy is at ceiling. Attention and saliency outputs are descriptive, and attribution methods of this family can appear plausible without reflecting the computation they purport to describe (Adebayo et al., 2018). Finally, the reproduction record for the experiments is incomplete in the manuscript as first written: model hyperparameters, framework versions, random seeds, and the code that produced each figure need to be recorded before the results can be independently reproduced. Appendix B lists the items required.

8 Conclusion

This paper reports a pipeline that takes continuous ICU EEG from raw signal to a classified, localized, and attributed output in one pass, and it reports the pipeline's behaviour on controlled synthetic data. Classification was errorless on held-out patients, localization recovered generators consistent with the class definitions, and attention was distributed across scalp regions rather than concentrated in one. Those outcomes validate the integration and are consistent with the construction of the data; they are not evidence of clinical capability, and the pipeline's calibrated certainty is demonstrably poor. The contribution is a working, documented, and honestly bounded system, together with the precise protocol for moving it onto the labelled corpus that is openly available for this task.

A Condensed Glossary

EEG. Recording of electrical activity at the scalp, produced mainly by synchronized pyramidal neurons in the cortex. **Montage.** The arrangement of electrodes and the pairs combined to form channels; nineteen electrodes in the standard configuration are the usual ICU arrangement. **Bipolar chain.** A line of electrodes in which each channel is the difference between adjacent sites, which suppresses distant activity and sharpens local detail. **Spectrogram.** A time-frequency representation showing how power at each frequency changes over the recording. **Artifact.** Signal of non-cerebral origin, including eye blinks, muscle activity, and mains interference. **Seizure.** Sustained abnormal rhythmic activity that evolves in frequency and distribution. **LPD and GPD.** Lateralized and generalized periodic discharges: repetitive discharges at a fixed rate, confined to one hemisphere or bilateral. **LRDA and GRDA.** Lateralized and generalized rhythmic delta activity, similarly distinguished by lateralization. **Forward model.** Mapping from cortical sources to scalp sensors through head geometry. **sLORETA.** An inverse method that estimates cortical current density and standardizes it to reduce localization bias. **Saliency map.** A visualization of which parts of an input most influenced a model's output. **Expected calibration error.** A summary of the gap between stated confidence and observed accuracy. **Soft targets.** Training against a distribution over categories rather than one label, which fits annotator disagreement.

B Reproduction Checklist

The following items must be recorded for the results to be independently reproducible, and are absent from the manuscript in its first version. *Data and splitting:* the exact recording identifiers used for background activity, the generator parameter values, the random seeds used to place and phase each generated source, and the patient identifiers assigned to each partition. *Model:* branch widths, kernel sizes, attention head count, layer counts, dropout, batch size, optimiser, learning rate schedule, epoch counts, and early-stopping criteria, with framework and library versions. *Training:* the construction of the soft target for each window, the loss implementation, and the treatment of class

imbalance. *Localization:* forward model specification, electrode coordinates, regularization parameter for the inverse solution, and atlas version. *Calibration:* the exact definition of the expected calibration error, the number of bins, and whether it is computed over maximum-probability predictions or per-class soft targets. *Artefacts:* the repository commit hash for the code behind every figure, and the released weights.

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